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SECR-1013

DIGITAL LOGIC

MINI PROJECT REPORT

(ELEVATOR CONTROLLER SYSTEM)

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BACKGROUND/OVERVIEW

In the realm of interactive applications, the need for a dynamic counter that facilitates both incremental and decremental counting has become increasingly apparent. The Counter Up and Down project aims to implement knowledge from this digital logic course by simulating an imaginary case study.

Counters are essential tools in many fields, from basic math problems to sophisticated applications in financial services, software development, and other fields. Conventional counters usually only make counting easier in the upward direction. But real-world situations frequently ask for the ability to count both up and down, providing a more complete and flexible answer.

The goal of the Counter Up and Down project is to offer a powerful, user-friendly tool that surpasses the constraints of traditional counters while also acknowledging the variety of user needs. Strategies that were used are group self-explore, creative problem solution and design and good reporting and presentation.

PROBLEM STATEMENT

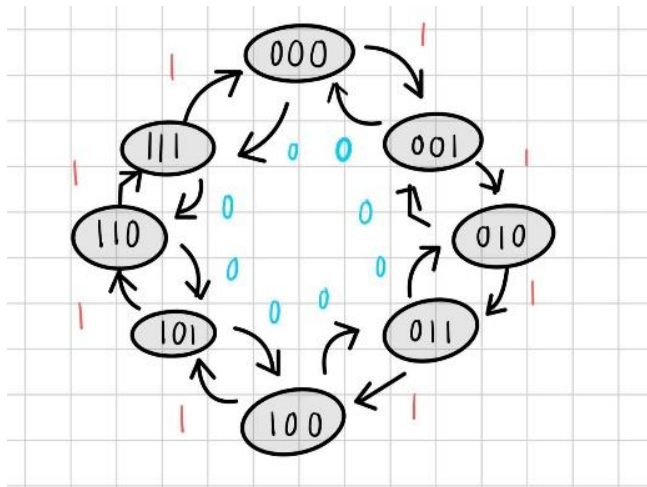
Elevator systems are essential to the vertical transportation realm because they facilitate safe and effective movement within structures. With the addition of a dynamic counting function that makes precise floor monitoring and control possible, the Elevator Controller System with Up and Down Counting Capability project aims to improve conventional elevator control mechanisms.

The case study provided:

- 1) To design an electronic controller for elevator/lift in a hotel building
- 2) A hotel visitor will use the elevator to go up/down from one level to another.
- 3) The visitor will key in the desired level to go, tag the visitor room card, close the elevator door, and push UP/DOWN button.
- 4) The elevator will go UP/DOWN by counting UP/DOWN one level to another until it reaches to desired level.

SOLUTION

STATE DIAGRAM:



EXPLANATION:

Count direction = 1

Count down direction = 0

STATE AND TRANSITION TABLE

x	PRESENT			NEXT			J ₂	K ₂	J ₁	K ₁	J ₀	K ₀
	Q _{2n}	Q _{1n}	Q _{0n}	Q _{2n+1}	Q _{1n+1}	Q _{0n+1}						
0	0	0	0	1	1	1	1	X	1	X	1	X
0	0	0	1	0	0	0	0	X	0	X	X	1
0	0	1	0	0	0	0	0	X	X	1	1	X
0	0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	0	0	1	1	X	1	1	X	1	X
0	1	0	1	1	0	0	X	0	0	X	X	1
0	1	1	0	1	0	1	X	0	X	1	1	X
0	1	1	1	1	0	0	X	0	X	0	X	1
1	0	0	0	0	0	0	0	X	0	X	1	X
1	0	0	1	0	1	0	0	X	1	X	X	1
1	0	1	0	0	1	1	0	X	X	0	1	X
1	0	1	1	1	0	0	0	X	X	1	X	1
1	1	0	0	1	0	0	1	X	0	0	X	1
1	1	0	1	0	1	1	1	X	0	1	X	1
1	1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	1	0	0	0	X	1	X	1	X	1

EXPLANATION

The excitation table shows the present state and the next state for three (3) outputs which are Q0, Q1, and Q2 as well as column for the input(X). Input 1=count up, whilst input 0=count down. The difference in the present state and the next state of an output gives us the values of J and K for each output.

The K-Map gives us a simplified Boolean expression to be used as input for each J and K in every flip-flop.

K-MAPS

J2		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	1	0	0	0
	01	X	X	X	X
	11	X	X	X	X
	10	0	0	1	0
		$J_2 = \bar{x} \bar{q}_1 \bar{q}_0 + x q_1 q_0$			

K2		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	X	X	X	X
	01	1	0	0	0
	11	0	0	1	0
	10	X	X	X	X
		$K_2 = \bar{x} \bar{q}_1 \bar{q}_0 + x q_1 q_0$			

J1		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	1	0	X	X
	01	1	0	X	X
	11	0	1	X	X
	10	0	1	X	X
		$J_1 = \bar{x} \bar{q}_0 + x q_0$			

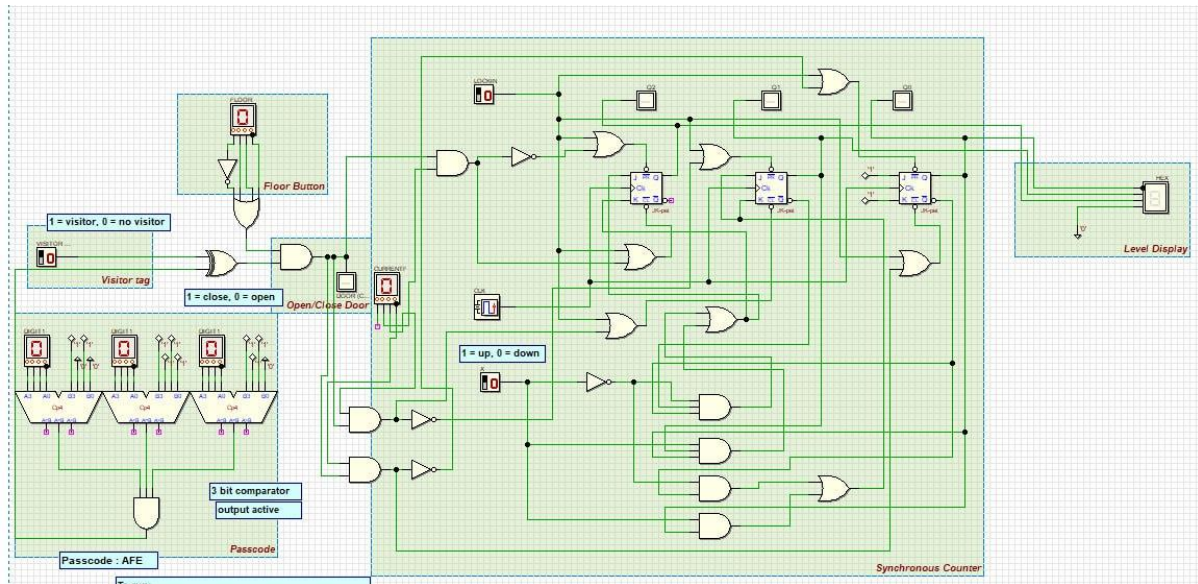
K1		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	X	X	0	1
	01	X	X	0	1
	11	X	X	1	0
	10	X	X	1	0
		$K_1 = \bar{x} \bar{q}_0 + x q_0$			

J0		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	1	X	X	1
	01	1	X	X	1
	11	1	X	X	1
	10	1	X	X	1
		$J_0 = 1$			

K0		$q_1 q_0$			
		00	01	11	10
$x q_2$	00	X	1	1	X
	01	X	1	1	X
	11	X	1	1	X
	10	X	1	1	X
		$K_0 = 1$			

SYSTEM IMPLEMENTATION

DEED DRAWING



EXPLANATION

1. Tag the visitor card to set VISITOR = 1 or use passcode = AFE (Visitor tag / Passcode)
2. Choose destined floor (Floor Button)
3. Choose current floor (CURRENTF)
4. Lock in the current floor (LOCKIN)
5. Set $X = 0$ for going down or set $X = 1$ for going up (X)
6. Press the UP/DOWN button to go UP/DOWN (CLK)
7. When you have at destined floor, tag the visitor card again to set VISITOR = 0 or assign passcode incorrectly to open the elevator gate (Visitor tag / Passcode)
8. To go to another level from the current floor, repeat steps 1-7

CONCLUSION

To sum up, the successful implementation of the counter up and down project not only met its intended objectives which is to simulate an imaginary case study by designing an electronic controller for elevator in a hotel building but also provided valuable insights into user-friendly design and the iterative process of development, paving the way for future enhancements and applications in diverse contexts.

The counter up and down emphasises how crucial it is to have thorough planning, strong safety precautions, and user-centric design when designing an electronic controller for elevator. The conversion of the counter up and down project to the electronic controller design for lifts, in short, necessitates a comprehensive strategy that strikes a balance between safety, usability, security, and maintenance concerns. By taking a thorough approach to these issues, we may develop electronic controllers that prioritise passenger safety and well-being while also exceeding regulatory requirements and improving overall user experience.

REFLECTION

The Deed Simulator is used as a simulation for the case study, and it is implemented very well. The challenges that were faced during the implementation were the misfilled the state and transition table, the usage of wrong gates and the lack of knowledge in Basic Design in Deed Simulator. But these challenges were discussed very well during the meeting and we overcame the issues before the due of the project. The solutions were included debugging, overcoming technical issues, and adapting to changing requirements.

By handling this case study, we also learned many valuable lessons. First, we got to improve and polish our skills in Digital Logic such as the problem-solving skills and critical and design thinking skills. Next, we also improved our knowledge on how to use the Deed Simulator and each usage of the gates. Lastly, we got to strengthen our teamwork because as we did a meeting to divide the task, we depend on each other and that lead to improvise the teamwork as well as communication skills. We also had the opportunity to improve and add our knowledge more about the Counter Up and Down as we've learned in the class.

ACHIEVEMENT

We achieved the main objectives:

1. To implement the Counter Up and Down by simulating the electronic controller for elevator/lift in a hotel building.
2. To construct the counter circuit using the Deed Simulator.
3. Introduce more features to input controller.
 - a) To detect visitor card hotel
 - b) To indicate door open/close
 - c) To use passcode

WEAKNESSES

As we handle the project, we identified the weakness that can help guide improvements and refinements. First, the counter may lack advanced features that could enhance its utility. For instance, it might only support basic counting without additional functionalities such as timers, custom intervals, or the ability to perform calculations. Next, inadequate testing of the electronic controller before deployment can lead to undiscovered bugs or flaws. Rigorous testing, including simulation of various scenarios and stress testing, is essential to identify and address potential issues. Lastly, if the design does not account for ease of maintenance and updates, the electronic controller may become outdated or difficult to service over time, leading to increased downtime and operational costs.

STRENGTHS

While the case study's main goal is not directly related to creating an electronic lift controller, some of its advantages can be applied in this situation. First, the counter project demonstrates efficient and well-organized design circuit. Clean and optimized design circuit contributes to the reliability and performance of the controller. Next, the counter project is designed to handle various scales or sizes of counting, this scalability concept can be beneficial in the context of elevator systems. The electronic controller is scalable to accommodate different elevator sizes and capacities.

IMPROVEMENTS

There are some enhancements that can be made to improve more about the Counter Up and Down project by designing the input controller for elevator. Improving the circuit design of the input controller for elevators involves incorporating advanced technologies, enhancing reliability, and ensuring efficient operation.

First, integrate more realistic simulation components that accurately represent the behaviour of physical hardware. This includes mimicking the characteristics of microcontrollers, sensors, and actuators to provide a more authentic simulation experience. Next, enhance the counter algorithm to include intelligent floor prediction based on historical usage patterns. This predictive feature can reduce waiting times by anticipating user destination floors. Lastly, enhance the circuit design with fault-tolerant features to simulate the response to sensor failures or communication errors. This can include redundant sensors and error-handling mechanisms within the counter logic.

REFERENCES

1. Website: Digital Electronic Deeds (2023) by Giuliano Donzellini, <https://www.digitalelectronicsdeeds.com/>.
2. Website: Electronics Tutorial (2024) , https://www.electronics-tutorials.ws/counter/count_4.html

APPENDICES

